

FCU3

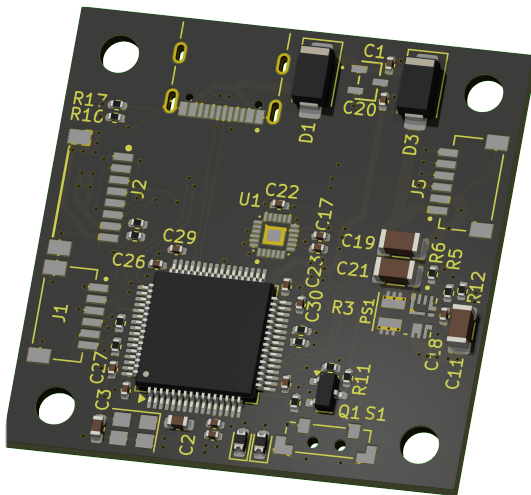
Variant: CHECKED

2025-12-11

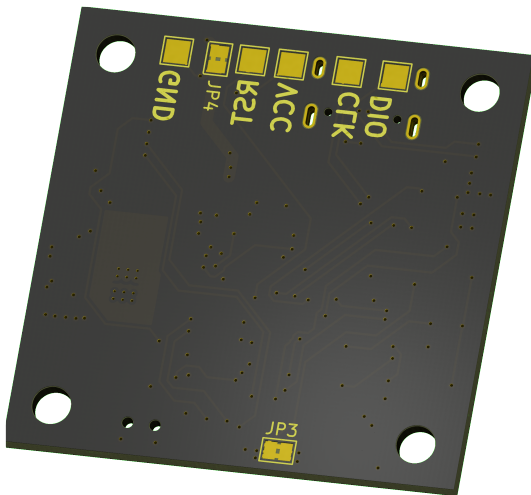
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 11-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company:	Variant:	Git Hash:
		Kaldstrand Electronics	CHECKED	4dadb50
	Sheet Title:	Board Name:	Project Name:	
		FCU3	Flight Controller 3	
	Sheet Path:	File Name:	Designer:	Date:
	/	FCU3.kicad_sch	Kaldstrand	2025-01-12
		Reviewer:	Size:	Revision:
			A3	+ (Unreleased)
				Sheet:
				1 of 8

[2] Block Diagram

Page #

+??V

+??V

+??V

Block Description

Block Description

Page #

SECTION

BLOCK A

Port Name

???

Page #

BLOCK B
BLOCK C

Port Name

???

???

Port Name

Page #

BLOCK D

Port Name

???

???

Port Name

Page #

BLOCK E
BLOCK F

Port Name

???

???

Port Name

Section

Page #

SECTION

BLOCK G

BLOCK H

BLOCK I

BLOCK J

Target specifications:

Input voltage:

?? - ?? V

Spec 2

??

Spec 3

??

Spec 4

??

Comments:

Company:

Kaldstrand Electronics

Variant:

CHECKED

Git Hash:

4dad50

Board Name:

FCU3

Project Name:

Flight Controller 3

Sheet Title:

Block Diagram

File Name:

Block Diagram.kicad_sch

Designer:

Kaldstrand

Date:

2025-01-12

Revision:

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Sheet Path:

/Block Diagram/

Reviewer:

Size:

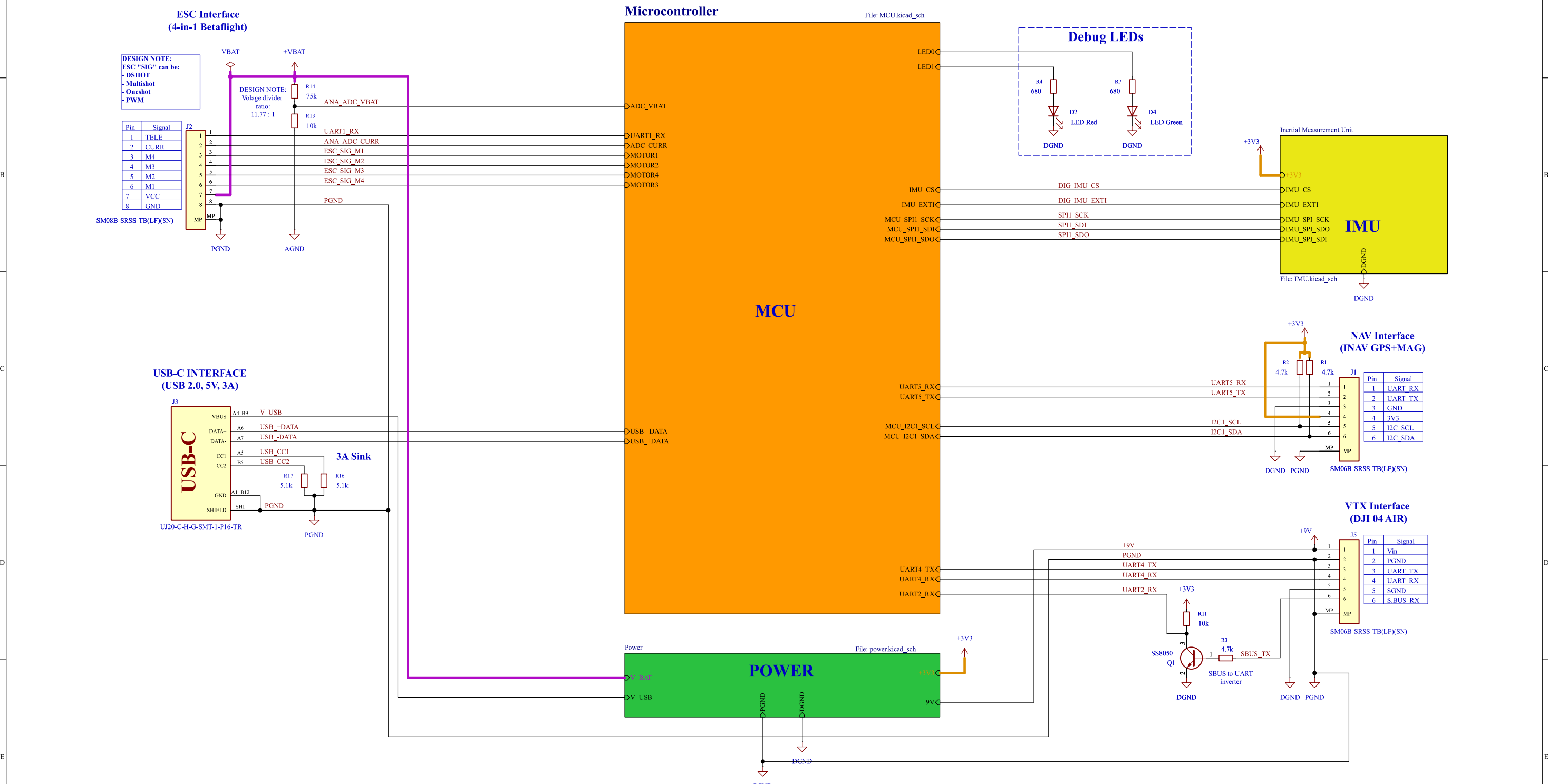
A3

Sheet:

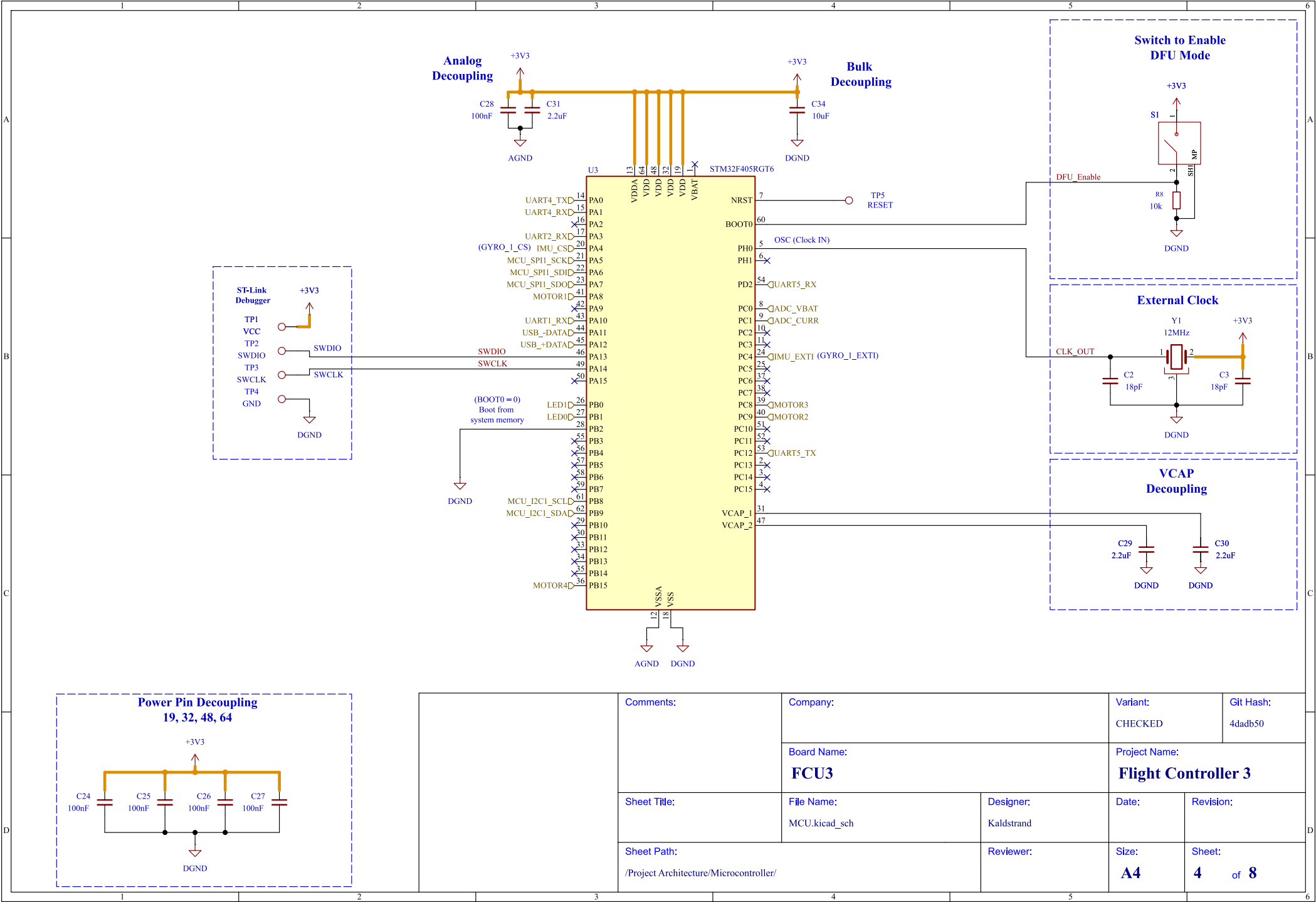
2 of 8

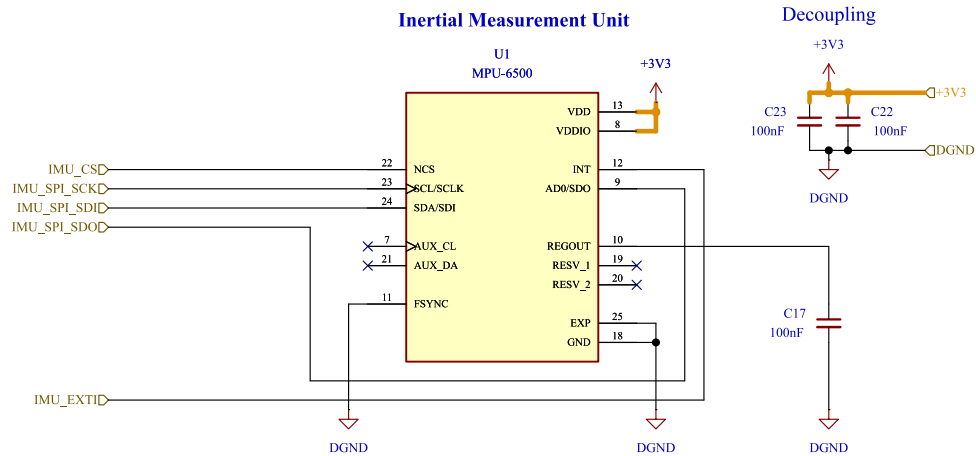
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

[3] Project Architecture

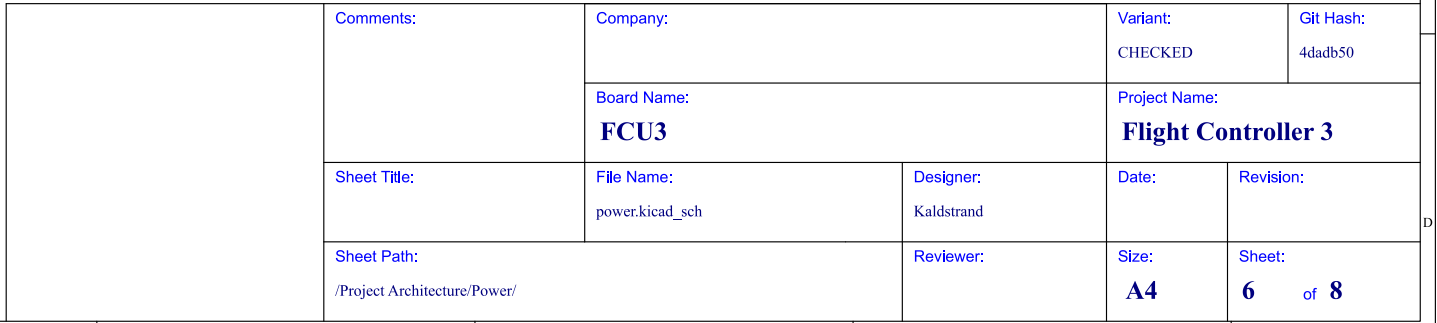


Grounding Strategy Ground Ties DGND JP3 AGND DGND AGND Mounting Holes H1 H2 H3 H4	PGND			Comments:		Company: Kaldstrand Electronics		Variant: CHECKED		Git Hash: 4dad50	
	Sheet Title: Project Architecture			Board Name: FCU3		Project Name: Flight Controller 3					
	Sheet Path: /Project Architecture/			File Name: Project Architecture.kicad_sch		Designer: Kaldstrand		Date: 2025-01-12		Revision: + (Unreleased)	
						Reviewer:		Size: A3		Sheet: 3 of 8	





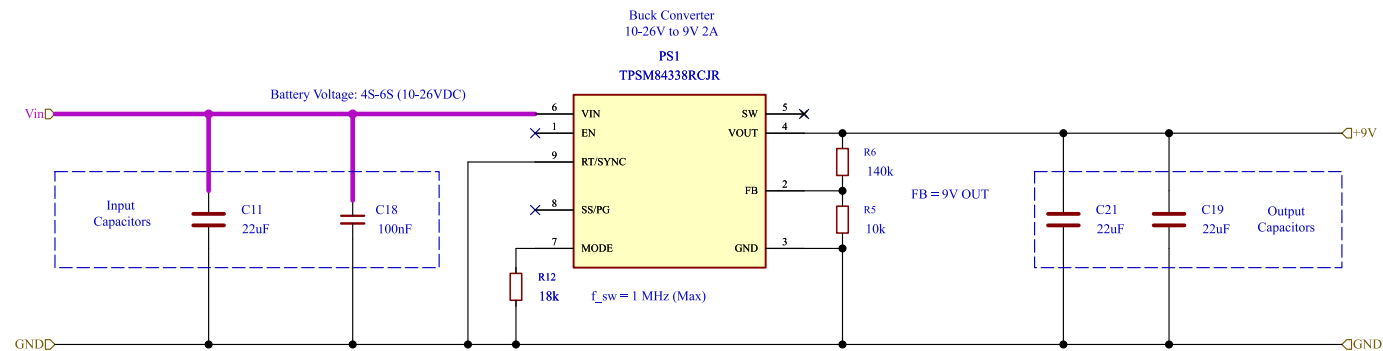
	Comments:	Company:	Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3	Project Name: Flight Controller 3	
	Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date: Revision:
	Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: A4 Sheet: 5 of 8



DESIGN NOTE:
 $R_{FBB} = (V_{OUT} - V_{REF}) / V_{REF} * R_{FBB}$
 $R_{FBB} = (9V - 0.6V) / 0.6V * 10k$
 $R_{FBB} = 140k$

 $V_{OUT} = 9V, V_{REF} = 0.6V, R11 = 10k, R6 = 140k$

To-Do:
Add design notes
explaining DC/DC settings.



	Comments:	Company:		Variant:	Git Hash:
				CHECKED	4dad50
		Board Name:		Project Name:	
		FCU3		Flight Controller 3	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
		stepdown_10-26vin_9v2a.kicad_sch	Kaldstrand		
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Power/10-26V to 9V2A Step-Down Converter/		A4	8 of 8	

[9] Revision History

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dad50
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8